

## CSCE 642 DEEP REINFORCEMENT LEARNING · MODULE 12 · LECTURE NOTES

# Reproducibility

*The variance is larger than the effects.*

## MODULE OUTCOMES

- Explain seed variance and its magnitude in this field.
- Explain why single-seed comparisons are uninformative.
- Report results with appropriate statistics.
- Explain the confounds beyond seeds.
- Read the literature with calibrated scepticism.

## 1 The finding

### SEED VARIANCE ROUTINELY EXCEEDS THE DIFFERENCES BETWEEN METHODS

**Run the identical algorithm, on the identical task, with the identical hyperparameters, varying only the random seed, and the results differ enormously** — frequently by more than the reported gap between two published methods.

**Two groups of five seeds drawn from the same algorithm can appear to be two different algorithms**, with non-overlapping confidence intervals and a clear apparent winner. **This has been demonstrated directly**, by splitting one algorithm's runs into two groups and comparing them as if they were different methods.

**So a comparison on one or two seeds carries almost no information about which method is better**, and a substantial portion of the published comparison literature is at that standard — which means a reader cannot distinguish a real advance from a favourable seed.

**This is measured, replicated, and not seriously disputed. It is a property of the field rather than an accusation against anyone** — the variance is caused by the problem's structure (§1's list), not by carelessness — **and knowing it is necessary to read the literature usefully at all.**

- **The data depends on the policy, which depends on the data. An early lucky trajectory leads to a better policy, which collects better data, which compounds** — and an early unlucky one compounds the same way downward. **This feedback is the root cause and is the one thing that genuinely distinguishes this from supervised learning's much smaller seed variance.**
- **Exploration is stochastic**, so two runs genuinely see different parts of the state space — not noisier views of the same data, but different data (Module 02 §3's corridor, where the seed decides whether the reward is ever found).

- **Instability** (Module 04 §1) means a run can diverge for reasons that are seed-dependent, so some fraction of runs fail outright while others succeed.
- **And a performance collapse is frequently unrecoverable** (Module 06 §2), so **outcomes are bimodal rather than spread**: some runs learn the task and some never do.
- **Which is why the mean is a poor summary. Averaging a bimodal distribution produces a number that describes neither mode**, and the standard deviation implies a shape the data does not have — hence §2's recommendations.

## 2 Reporting

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MINIMUM: FIVE SEEDS. Ten if you can afford it. And REPORT HOW MANY -- a result with the seed count omitted cannot be assessed at all.

PLOT ALL THE RUNS, or a median with an interquartile range. NOT a mean with a standard deviation, which assumes a distribution the data does not have (section 1).

REPORT THE INTERQUARTILE MEAN across seeds and tasks. It is robust to the outlier runs that dominate a plain mean, and it is what the Rliable tooling computes for you.

REPORT PERFORMANCE PROFILES: the fraction of runs exceeding each score threshold. This shows the whole distribution rather than one statistic, and it makes bimodality visible immediately.

REPORT THE SAMPLE BUDGET. "Better after 10M steps" and "better after 1M steps" are different claims, and the learning curves frequently cross.

TUNE THE BASELINE AS HARD AS YOU TUNED YOUR METHOD. An untuned baseline is not a comparison. This is the single largest source of overstated improvements in the field, and it is CSCE 633 Module 02's baseline discipline with a very specific target.

AND FIX EVERY SOURCE OF RANDOMNESS you can control: environment seeds, network initialisation, action sampling, and the framework's global RNG state. Then report what you fixed, because full determinism is usually unattainable.

**The interquartile mean and performance profiles are the field's own recommended practice**, with free tooling that computes them and produces the plots — **so adopting them costs essentially nothing** and immediately puts your results above the median standard. **Giving the concrete standard matters**, because vague calls for rigour do not change what people do and a specific checklist does.

### 3 The other confounds

| Confound                     | Effect                                                                                                                       | What to do about it                                                                                           |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| Implementation details       | Can exceed the algorithmic difference entirely (Module 06 §3's finding).                                                     | Match the details, or run both methods in one codebase — which is what CleanRL and similar efforts exist for. |
| Hyperparameter tuning effort | Whoever tuned more wins, independently of the method.                                                                        | Give both methods an equal and reported search budget.                                                        |
| Environment version          | Reward scales, termination conditions, and observation definitions change between library versions, sometimes substantially. | State the exact version, and never compare numbers across versions.                                           |
| Evaluation protocol          | Best checkpoint, final policy, and average over training give very different numbers.                                        | State which you used, and prefer the final policy or an average over the last portion of training.            |
| Selective task reporting     | Reporting the subset of tasks where your method wins.                                                                        | Report the whole benchmark suite, including the tasks where you lose.                                         |

'Best checkpoint over training' deserves special mention: it is a form of test-set selection (CSCE 633 Module 03 §3's leakage), because the checkpoint was chosen using the evaluation you are then reporting. It is extremely common in this field and it inflates every number it touches — and with §1's variance, selecting the best of fifty checkpoints can manufacture an improvement from nothing.

## 4 Reading the literature

- **How many seeds, and is the spread shown? If fewer than five, or if only a mean is reported, the comparison is weak evidence** whatever the stated improvement.
- **Was the baseline tuned, by whom, and with what budget?** A baseline taken from another paper's reported number, run on a different codebase, is not a controlled comparison.
- **Is the claimed improvement larger than the seed spread? If not, it is not established** — and this single check disposes of a surprising number of results.
- **Is there a released implementation that reproduces the reported number? Frequently there is not,** and 'code available' is not the same as 'code reproduces the figures'.
- **And does the method appear in anyone else's benchmarks a year later? Durable adoption by people with no stake in it is the strongest available evidence, and it is the most reliable single filter** — it costs one search, and methods that genuinely work get used while methods that do not quietly disappear.

### WHY THIS MODULE EXISTS HERE AND NOT AS A FOOTNOTE

This course cannot teach you to evaluate reinforcement learning results without stating plainly that the field's standard practice has been inadequate — and it would be a disservice to teach the algorithms and omit this.

**It is improving.** The statistical tooling exists and is free, the variance measurements are published and well-cited, and major venues increasingly require seed counts and spread reporting.

**And it is still common to encounter three-seed comparisons with confident claims,** which §1's measurements say cannot be supported — so the calibrated response when reading is scepticism about the *specific comparison* rather than cynicism about the field.

**So the practical instruction is simple: five seeds minimum, spread shown, baseline tuned equally, protocol and sample budget stated. Meet that standard and your results mean something regardless of what anyone else does** — which is this program's position throughout, and here it has unusually concrete numbers behind it.

## Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

| Resource                                                                 | What it covers for this module                                                                                                 |
|--------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Henderson et al. — Deep Reinforcement Learning That Matters (free)       | §1's <b>measurements</b> , including the same-algorithm split-group demonstration. Read this before trusting any comparison.   |
| Agarwal et al. — Deep RL at the Edge of the Statistical Precipice (free) | §2's <b>recommended statistics</b> , with the free Rliable library that computes interquartile means and performance profiles. |

| Resource                                                                               | What it covers for this module                                                                   |
|----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| Engstrom et al. — Implementation Matters in Deep Policy Gradients (free)               | <b>§3's first row, measured</b> — the implementation confound isolated from the algorithmic one. |
| Jordan et al. — Evaluating the Performance of Reinforcement Learning Algorithms (free) | <b>A complete evaluation protocol</b> , including the hyperparameter-effort confound of §3.      |

## Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **Run one algorithm with ten seeds** on one task and plot all ten learning curves.
- 2 **Split them into two groups of five** and compare the groups as if they were different methods. Report what you would have concluded.
- 3 **Compute the mean, median, and interquartile mean** and report how much they differ.
- 4 **Plot a performance profile** and identify whether the distribution is bimodal.
- 5 **Compare an untuned baseline against your tuned method**, then tune the baseline equally and compare again.
- 6 **Report the best checkpoint and the final policy** for the same runs, and report the inflation.
- 7 **Run the same code on two environment library versions** and compare.
- 8 **Take a published result** and apply §4's five questions to it. Report your assessment.
- 9 **Install Rliable** and reproduce its plots for your own results.
- 10 **Write your project's results section** to §2's standard.

## Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 State the seed-variance finding and its magnitude.
- 2 What can two groups of five seeds from one algorithm look like?
- 3 Give four causes of the variance and identify the root one.
- 4 Why is a mean with a standard deviation the wrong summary?
- 5 Give the seven items in the reporting standard.
- 6 Name five confounds beyond seeds.
- 7 Why is 'best checkpoint' a leakage problem?

- 8 Give five questions to ask of a published result.
- 9 Which is the most reliable single filter, and why?